DIVIDE AND CONQUER: MERGESORT

Link to practice handout

https://bit.ly/Divide-and-Conquer-Practice

Divide and Conquer Algorithms

Algorithm Approach:

- **Divide** a large problem into sub-problems
- **Solve** each sub-problem
- **Combine** the solutions of sub-problems to obtain the solution for the original problem

Merge Sort Algorithm

```
MergeSort(vector v)

• Divide v into left half and right half

• Sort the left half, then sort the right half

• Combine (merge) the two sorted halves
```

Example run of mergesort

[7 2]

Example run of mergesort

[7 2 3 -1]

[7 2 5 3 -1]

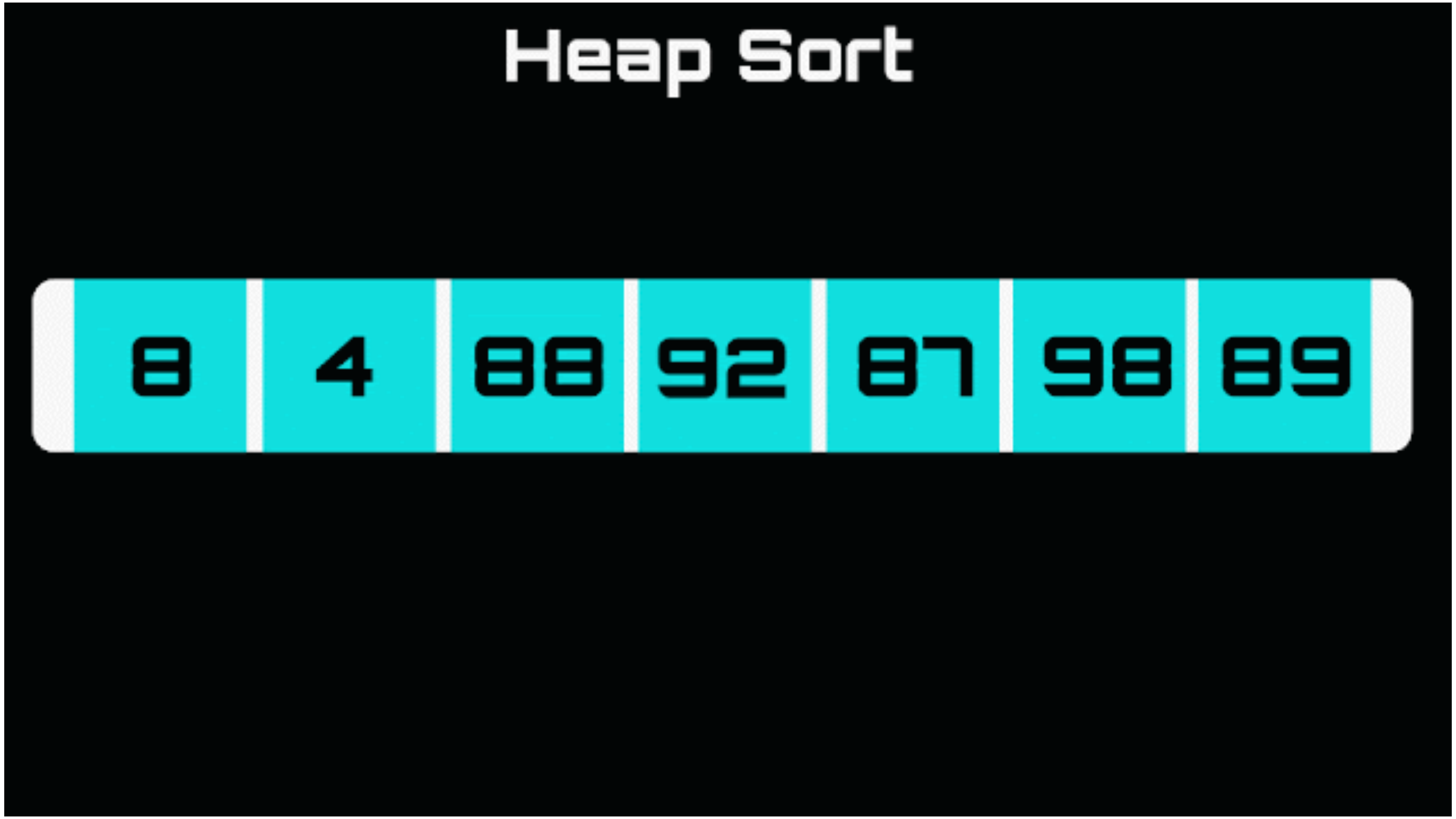
Activity 1: Trace merge sort on the above example input with 5 elements
Draw the final binary tree representation of the trace.
What is the height of the resulting binary tree?

Time Complexity of Mergesort

Space Complexity of Mergesort

What is the space complexity of mergesort?

A. O(n)
B. O(n log n)
C. O(2^n)
D. Something else



Source: https://cosmictechie.hashnode.dev/heap-sort

HeapSort Algorithm

```
HeapSort(vector v)
    for i from n/2-1 down to 0:    // heapify: build max heap
      bubble_down(v, n, i)

    for i from n-1 down to 1:      // Invariant:
      swap(v[0], v[i])                   heap: v[0..i-1]
      bubble_down(v, i, 0)               sorted: v[i..n-1]
```

Heapsort: https://visualgo.net/en/heap

Why Mergesort over Heapsort?

- Same O(n log n) time, but heapsort has poor cache performance - why?
- Mergesort is **stable**: equal elements keep their original relative order
- Mergesort parallelizes naturally